Mitacs GRI Professor Matching Guide

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Purpose: To help you convert a first-phase GRI shortlisting into a professor match and a final Mitacs offer.

This guide is based on the projects listed in the application profile you shared (Application ID 252214), your CV and transcript in that same file, and published Mitacs matching timelines. It is **not** an official Mitacs document, and it does not invent any professor's lab details, interview date, or project description beyond the project titles you applied for. Always follow the instructions in your Mitacs portal and in any email from a professor or from Mitacs itself, even if they differ from this guide.

First, the most important distinction. First-phase selection means your application has passed screening and is going to professors for matching. It is not yet a final offer. A final offer comes only after professors review, interview and rank candidates, and Mitacs finalises the match. Treat every professor contact from now on as part of the selection, not as a formality.

1. How matching works, in plain words

After student applications close, Mitacs vets them for eligibility. Then, in the matching window (November to early February for the 2027 cycle, according to university Mitacs pages), professors receive a shortlist for their project. They review applications, may interview candidates, and rank them in order of suitability. Mitacs collects the professor and student rankings and finalises offers. Final results are announced in February.

That means three things for you:

- **Professors choose, Mitacs confirms.** A friendly interview is a good sign, but it is not a guarantee until Mitacs sends the formal offer.
- **You may hear from more than one professor, or from none for a while.** Silence in November does not mean rejection. Some professors contact candidates in November or December; others decide later in the window.
- **Your ranking matters too.** If your portal offers a chance to re-order your project list during matching, rank by true research fit, not by university name or city. A professor can tell very quickly whether you chose their project for its topic or for its location.

2. Your 10 projects, ranked by fit

The table below uses only the project titles, provinces and professor names from your application. Fit is judged against your FYP (low-head hydropower on KP canals), your Tarbela experience, your floating-PV work, Energiscope, and your stated skills. It is a preparation guide, not a prediction of which professor will contact you.

Rank	Province /	Project title	Fit	Why, in one line

	Professor			
1	Alberta — Bo Zhang	Hybrid Advanced Geothermal–Hydropower Systems for Reliable Low-Carbon Heat and Electricity in Canada	**High**	Closest to hydropower + hybrid systems; geothermal heat is the new part to learn.
2	Ontario — Mohsin Jamil	Techno-Economic Design and Dynamic Analysis of a Hybrid Renewable Energy System for a Remote Community Using HOMER Pro and MATLAB/Simulink	**High**	Almost a direct match to your Charsadda PV-hydro-battery case study.
3	Québec — Akhtar Hussain	AI-Assisted Resilient Operation of Microgrids Under Component and Grid Outages	**Medium-High**	Fits microgrid monitoring, but expect AI questions; do not overclaim ML.
4	Nova Scotia — Adel Merabet	Energy monitoring for renewable energy system	**Medium-High**	Fits Energiscope and energy monitoring, if you can show data work.
5	Newfoundland and Labrador — Iman Nosoohi	Government Interventions for Energy Efficient AI Technologies: A Game-Theoretic Analysis	**Medium-Low**	Policy + game theory + AI; further from field hydropower.
6	Québec — Yili Tang	Carbon neural transportation and emission analyses	**Medium-Low**	Carbon/emissions analysis; title is broad, so ask for the project scope if contacted.
7	Ontario — Omar Alam	ML-Driven Approach for the Electric Vehicle Routing Problem	**Low-Medium**	ML + optimisation + EVs; you would need a strong ML/optimisation story.
8	British Columbia — Haibo Feng	Predicting the energy performance and energy use of urban buildings in Vancouver	**Low-Medium**	Buildings energy, not hydropower; data modelling is the bridge.
9	Alberta — Yize	Interconnecting AI Data Centers	**Low-Medium**	Power grids are relevant,

	Chen	to Power Grids		but data-centre load is a new domain.
10	Québec — Andreas Ruediger	Photocatalysis for clean hydrogen production	**Low**	Hydrogen is in your coursework, but photocatalysis is chemistry-heavy and far from your field work.

Preparation priority: Spend 70% of your preparation time on ranks 1 and 2, 20% on ranks 3 and 4, and only a quick one-page note each for ranks 5–10, so you are not caught cold if one of them contacts you first.

3. What professors are actually judging

From past GRI matching reports, professor interviews are usually friendly discussions, not formal exams, but they do test fit. Professors are judging, in roughly this order:

1. **Research fit:** Have you read the project title carefully, and can you explain why this project, in your own words?
2. **Evidence, not adjectives:** Can you name something you actually did — a site, a dataset, a calculation, a tool, a result — rather than saying “I am passionate about renewable energy”?
3. **Honest skill level:** Can you say what you can do, what you are learning, and what you would need help with, without exaggerating?
4. **Reliability:** Will you reply promptly, turn up on time across time zones, work independently for 12 weeks, and finish a deliverable?
5. **Communication:** Can you explain a technical idea clearly in English, and ask a sensible question back?

Some professors also give a small task instead of, or as well as, a call — for example, summarising a paper and saying how you would approach the problem. Treat any task as part of the interview, and submit it on time, in your own words.

4. Your profile: strengths to lead with, gaps to handle honestly

Lead with these, with a concrete detail each time

- **FYP — low-head hydropower on KP irrigation canals:** 12 canal sites, data from 3 government departments, techno-economic comparison. Say the formula you use, $P = \rho g Q H \eta$, and explain that irrigation water comes first — that shows engineering judgement, not just enthusiasm.
- **Tarbela:** 4-week WAPDA Tarbela internship (turbines, generators, switchyard, operations) plus the Tarbela 5th Extension project internship. This is rare for an undergraduate and directly relevant to ranks 1 and 2.
- **Floating PV research:** 120 weekdays of field data (irradiance, module temperature, water temperature, output efficiency, wind speed), structured in a multi-sheet Excel workbook, analysis complete, manuscript in preparation. Do not claim it is published — “in preparation” is honest and still strong.
- **Energiscope:** A deployed web platform visualising 10+ years of NEPRA generation data, about 40 GW of capacity, into capacity-mix and output-trend views. This is your best proof of data work for ranks 3 and 4.
- **Grades trend:** CGPA 3.46/4.00 (86.5 in your application). If asked about the Spring 2024 dip to 2.76, answer briefly and forward-looking: you recovered to 3.81, 3.75 and 3.69 in later semesters. Do not over-explain or blame anyone.

Handle these gaps before a professor finds them

- **HOMER Pro:** Your CV says “in progress — target completion 10 September 2026”. That date has now passed. Before any interview, update your status honestly: what part is done (for example, the Charsadda load profile and solar resource assessment), what part is still in progress, and when you will finish it. For rank 2, this is the single most important update.
- **Python and GIS:** Your application lists Python and data literacy, and your CV lists MATLAB as working proficiency. In reality you are a beginner in Python and still developing GIS. Say that plainly, then show what you have actually done in Python (for example, the hydropower power calculation) and what you are learning next. A professor would rather hear “beginner, but I can do X” than discover an exaggeration in week one.
- **Machine learning:** Your CV mentions training an ML model on FYP site data, and you hold introductory ML/AI certificates. That is not the same as ML project experience. For ranks 3, 7 and 9, prepare to explain, in simple terms, what data you used, what the model was supposed to do, and where it is limited. Do not let a professor assume you are an ML specialist.

- **Graduation and availability:** Your application lists anticipated graduation in October 2027. Be consistent with that in every professor conversation, and be clear about a May–July start window. If your portal shows an older summer year in any template text, do not repeat that year to a professor — confirm the correct 2027 dates in your portal first.
- **English:** You have no English proficiency exam on file, which is acceptable for the Pakistan route, but it means the interview itself is your English test. Practise speaking your answers out loud, slowly and clearly, not just reading them.

5. Your 90-second introduction (adapt, do not memorise word for word)

> *"Hello, I am Muhammad Ismail, a final-year Energy Engineering student at UET Peshawar. My final-year project is a feasibility study of low-head hydropower on KP irrigation canals, assessing 12 sites with data from government departments, using power, head, flow and efficiency to compare sites, with irrigation as the first priority. I have field experience at Tarbela, including the 5th Extension project, and I have collected 120 days of floating solar field data for a benchmarking study that is now in preparation. I also built Energiscope, a platform that visualises over ten years of Pakistan's generation data. I am interested in your project because [one sentence linking the project title to your work], and I would like to learn [one specific skill or method from the project] while contributing [one concrete thing you can already do]."*

Fill the two bracketed parts differently for every professor. A generic introduction that could be sent to all ten professors is a weakness, not a time-saver.

6. Likely interview questions, with how to answer them

1. Tell me about yourself. Use the 90-second structure above. End with the project link, not with your hobbies.

2. Why this project? Name one idea in the title, link it to one piece of your work, and name one thing you want to learn. Example for rank 2: "Hybrid sizing for a remote community is exactly what I am doing in Charsadda with PV, hydro and battery, and I want to learn how to test that design dynamically in Simulink, not just size it statically."

3. Explain your FYP in two minutes. Problem (canal falls exist but irrigation comes first) → method (site data, head/flow, $P = \rho g Q H \eta$, techno-economic comparison) → what is done, what is still in progress, what you have learned. Do not invent a final power figure for a site you have not measured.

4. What did you do at Tarbela? Be specific: turbines, generators, switchyard, operations and load coordination in the 4-week internship, and electrical/mechanical exposure in the 5th Extension placement. If you did not do a particular task, say so.

5. What tools can you use? Answer in three levels: “I can do X confidently” (for example, Excel data work, AutoCAD basics, MATLAB at working level), “I am learning Y” (HOMER Pro, Python), “I have not used Z yet, but I can learn it in the first two weeks because…”. Never list a tool you cannot open and use.

6. Tell me about a time you handled a difficult dataset or field problem. Use the floating-PV data: 120 weekdays, multiple variables, missing or noisy readings if any, how you structured the workbook, and what the comparison showed. Keep it factual.

7. Why Canada, and why now? Keep it academic: a specific research area, a chance to work with a Canadian research group, and a step towards a research career. Do not make it about immigration or a future visa — that is not what a professor is selecting for.

8. Are you available for the full 12 weeks, and when can you start? Answer with your real availability, and confirm the start window in your portal before the call. If you have an exam or FYP deadline that clashes, raise it early, not after an offer.

9. What questions do you have for me? Always ask at least two. Good examples are in Section 8 below. “No questions” looks uninterested.

10. What would you do in the first two weeks? Show a plan: read the professor's key papers and project data, learn the required tool, reproduce a baseline calculation, and agree a deliverable and meeting rhythm. Professors love a candidate who has thought about week one.

7. Project-specific preparation sheets — ranks 1 and 2

Rank 1 — Alberta, Bo Zhang: Hybrid Advanced Geothermal–Hydropower Systems

Core idea to understand before the call: A hybrid system that combines geothermal heat with hydropower electricity, so that heat and power can support each other and make low-carbon supply more reliable than either source alone. You do not need to be a geothermal expert, but you should be able to explain, in simple terms, how geothermal heat is produced (hot water/steam from underground), how hydropower produces electricity (head, flow, efficiency), and why combining them can help when one source varies.

Revise: P = ρgQHη and what happens when head or flow changes; the difference between heat (thermal energy) and electricity; why “reliable” matters for a grid (matching supply to demand, not just total energy); and one question about how the hybrid is sized in Canada — for example, whether geothermal is used mainly for heat, for power, or for both in that project.

Likely questions: How does your canal hydropower work relate to a hybrid system? What do you know about geothermal? How would you compare two hybrid options — by cost, by emissions, by reliability, or by all three? Answer with your FYP techno-economic approach, and be honest that geothermal is new to you.

Starter plan to offer: Week 1: read project background and reproduce a simple hydro power calculation. Week 2: build a simple comparison table for hydro-only versus hybrid, using the project's own assumptions, and agree the next step with the professor.

Rank 2 — Ontario, Mohsin Jamil: Hybrid Renewable System for a Remote Community (HOMER Pro + MATLAB/Simulink)

Core idea to understand before the call: Sizing a hybrid system (for example, solar + hydro + battery, with or without diesel) for a community that is not well connected to the main grid, then testing whether that design actually works dynamically — voltages, frequency, load changes — in MATLAB/Simulink, not just on paper.

Revise: HOMER Pro workflow in order: load profile → resource data (solar, hydro) → component sizes and costs → simulation → techno-economic results (cost of energy, net present cost, renewable share). Simulink basics: what a dynamic simulation adds over a static sizing (transients, control, stability when load or solar changes suddenly). Your Charsadda case study is your bridge — prepare to walk through it step by step.

Likely questions: Walk me through your Charsadda hybrid design. What load did you assume, and why? What HOMER Pro results have you actually produced so far? What happens in Simulink if solar drops suddenly? Answer with real numbers from your case study if you have them, and say clearly what is still preliminary.

Starter plan to offer: Week 1: reproduce your Charsadda HOMER Pro model and document every assumption. Week 2: build a simple Simulink model of the same system and test one disturbance (for example, a load step), then discuss the result with the professor.

For ranks 3 and 4, prepare one page each, not a full sheet. Rank 3 (microgrids): revise

microgrid basics, what “resilient operation under outages” means, and one honest ML

sentence. Rank 4 (energy monitoring): revise what Energiscope actually measures and

visualises, and how you would monitor a renewable system (what sensors, what data, what

dashboard, what alarm or report).

8. Email and communication playbook

When a professor emails you

- Reply within 24 hours, even if only to thank them and propose times. Professors are contacting several candidates.
- Check your spam/junk folder every day during matching. Past GRI interns report professor emails landing in spam.
- Convert times carefully. Pakistan is PKT (UTC+5). Alberta is Mountain Time, Ontario and Québec are Eastern Time, Nova Scotia is Atlantic Time, Newfoundland is Newfoundland Time, and British Columbia is Pacific Time. Always write the time in both PKT and the professor's time zone, and confirm the date — a 9 a.m. call in Canada can be evening in Pakistan.
- Test your camera, microphone and internet before a video call, and have a quiet background. Join 5 minutes early.

Reply template (adapt, do not copy blindly)

> *Subject: Re: [Project title] — Mitacs GRI interview*
>
> *Dear Professor [Surname],*
>
> *Thank you very much for contacting me about your project, [project title]. I am very interested in this project because [one sentence linking it to your FYP or case study].*

> *I am available for a video call at the following times (PKT / your time): [give 3–4 options across different days]. Please let me know if another time would suit you better.*

> *Thank you again for considering my application.*

> *Kind regards,*
> *Muhammad Ismail*
> *BS Energy Engineering, UET Peshawar*

What not to do

- Do not mass-email all ten professors asking for a match. In GRI, professors contact shortlisted students through Mitacs; unsolicited mass emails can hurt you.
- Do not tell one professor you have already accepted another project unless Mitacs has sent you a formal offer.
- Do not exaggerate a skill, a grade, a publication, or a project role. Professors check, and Mitacs can terminate an application for misrepresentation.
- Do not ignore a professor because you prefer another project. Reply politely to everyone, and let Mitacs finalise the match.
- Do not discuss visa, immigration or future study plans as your main motivation in a professor interview. Keep the focus on the research.

Questions to ask a professor (pick two or three)

- Who will I work with day to day — you, a graduate student, or a research team?
- What does a successful 12-week deliverable look like for this project?
- What tools or data should I learn or read before I arrive?
- How often do you meet with interns, and how do you prefer to communicate?
- Is there a paper, dataset or report you would like me to read before we start?

9. Timeline and document checklist

When	What to do
Now — November	Update HOMER Pro status; prepare one-page notes for ranks 1–4; practise your 90-second introduction out loud; check email + spam + portal daily.

November — December	Be ready for professor interviews and small tasks. Reply within 24 hours. Keep a log: professor, project, date contacted, interview date, task deadline, outcome.
January — February	Matching continues; final results are announced in February. Do not assume silence means rejection until Mitacs notifies you.
February — April	If matched: pre-internship preparation with your host professor (reading, start/end dates, logistics) and Mitacs paperwork. Check passport validity and any visa/work-permit steps Mitacs instructs — do not book travel until Mitacs confirms.
May — July start	12 consecutive weeks in Canada, in person, starting between May 1 and July 31 (confirm your exact dates in your offer, not from this guide).

Documents to have ready, in one folder: passport, transcript, CV (updated HOMER Pro status), reference letter details, FYP one-page summary, Energiscope link/screenshots, floating-PV summary, and a list of your 10 projects with professor names and provinces. Do not send any of these to a professor unless they ask, or Mitacs instructs you to.

10. If matching does not go as hoped

- **If no professor contacts you by January:** This is common. Matching runs into February, and professors decide at different speeds. Keep checking the portal, keep preparing, and keep your FYP and master's outreach going in parallel — do not put everything on hold for one programme.
- **If you are matched to a lower-ranked project:** Judge it on the actual project description and professor, not only on your original rank. A good supervisor in a slightly different topic can be better than a perfect title with a poor fit.
- **If you are not matched at all:** It is not a judgement on your whole profile. GRI is highly competitive and matching depends on professor needs, funding and rankings. Your next best Canada step is the thesis-master's route: contact professors directly for a research master's, using the same FYP and field evidence, and consider a Mitacs Globalink Research Award while you are still enrolled, if timing allows.

11. One-page checklist (print this)

- ☐ Email + spam + Mitacs portal checked today
- ☐ HOMER Pro status updated honestly on CV
- ☐ 90-second introduction practised out loud

- ☐ One-page notes ready for ranks 1, 2, 3 and 4
- ☐ FYP explained in 2 minutes, with $P = \rho g Q H \eta$ and irrigation-first
- ☐ Tarbela, floating-PV and Energiscope stories ready, with numbers
- ☐ Two questions ready for every professor
- ☐ Time-zone conversion checked before every call
- ☐ Reply to any professor within 24 hours
- ☐ No skill, grade or publication exaggerated
- ☐ Interview log updated after every contact
- ☐ Backup plan (thesis master's outreach) still moving

Good luck, Ismail — you have already cleared the hardest filter, which is getting shortlisted. Now the work is preparation, prompt replies, and honest, specific answers. If a professor contacts you, send me their project title and any task they set, and we will prepare for that exact interview next.